

Unit-5

Environmental Protection; *Environmental Protection Act 1986, Initiatives by Non Governmental Organizations (NGO's), Human*
Human Population and the Environment: *Population growth, Environmental Education, Women Education.*

ECOLOGY RELATED LEGISLATIONS IN INDIA

Considering the complexity in terms of conservation of biodiversity, sustainability of the natural ecosystems and the livelihood dependence of the local communities, the government needs to address national and global issues related to carbon accumulation, biodiversity conservation, and continued flow of ecosystem services. These measures provide opportunities for strengthening documentation and data collection; empowering local communities by recognizing responsibilities, ownerships, rights, and concessions; and creating suitable institutions. The legislative provisions developed as a follow-up to such national policies are listed below:

THE ENVIRONMENT (PROTECTION) ACT 1986

Environment Protection Act, 1986 is an Act of the Parliament of India. It was enacted in May 1986 and came into force on 19 November 1986. It has 26 sections and 4 chapters. The Act is widely considered to have been a response to the Bhopal gas leak. The Act was passed by the Government of India under the Article 253 of the Constitution of India, which empowers to union government to enact laws to give effect to international agreements signed by the country.

The **Environment (Protection) Act (EPA)** was enacted in 1986 with the objective of providing the protection and improvement of the environment.

- It **empowers the Central Government** to establish authorities charged with the mandate of preventing environmental pollution in all its forms and to tackle specific environmental problems that are peculiar to different parts of the country.
- The Act is **one of the most comprehensive legislations** with a pretext to protection and improvement of the environment.

Salient Features of the EPA Act

- **Powers of the Central Government:** The Central Government shall have the power to take all such measures as it deems necessary or expedient for the purpose of protecting and improving the quality of the environment in coordination with the State Governments.
 - **Plan and Execute a nation-wide programme** for the prevention, control and abatement of environmental pollution.

- **Lay down standards for the quality of environment** in its various aspects.
- Lay down **standards for emission or discharge of environmental pollutants** from various sources.
- The **restriction of areas** in which any industries, operations or processes or class of industries, operations or processes shall/ shall not be carried out subject to certain safeguards.
- The central government as per the Act has the power to direct:
 - The closure, prohibition or regulation of any industry, operation or process.
 - The stoppage or regulation of the supply of electricity or water or any other service.
- **Restriction on Pollutant Discharge:** No individual or organisation shall discharge/emit or permit to discharge/emit any environmental pollutant in excess of the prescribed standards.
- **Compliance with Procedural Safeguards:** No individual shall handle or shall be caused to handle any hazardous substance except in accordance with the procedure and without complying with the safeguards, as prescribed.
- **Powers of Entry and Inspection:** Any person empowered by the Central Government shall have a right to enter (with the assistance deemed necessary) at any place:
 - For the inspection of compliance of any orders, notifications and directions given under the Act.
 - For the purpose of examining (and if required seizing) any equipment, industrial plant, record, register, document or any other material object may furnish evidence of the commission of an offence punishable under this Act.
- **Establishment of Environmental Laboratories:** The Central Government, as per the Act, is entitled to:
 - Establish environmental laboratories.
 - Recognise any laboratory or institute as environmental laboratories to carry out the functions entrusted to such a laboratory.
- **Appointment of Government Analyst:** A Government Analyst is appointed by the Central Government for the analysing the samples of air, water, soil or other substance sent to a recognised environmental laboratory.
- **Penalties for Offences:** Non-compliance or Contravention to any of the provisions of the Act is considered as an offence.

- Any offences under the EPA are punishable with the imprisonment of upto five years or a fine upto one lakh rupees or both.
- **Offences by Companies:** If an offence under this Act is committed by a company, every person directly in charge of the company, at the time of the commitment of offence, is deemed to be guilty unless proven otherwise.
- **Offences by Government Departments:** If an offence under this Act has been committed by any Department of Government, the Head of the Department (HoD) shall be deemed to be guilty of the offence

Drawbacks of the Act

- **Complete Centralisation of the Act:** A potential drawback of the Act could be its centralization. While such wide powers are provided to the Centre and no powers to the state governments, the former is liable to its arbitrariness and misuse.
- **No Public Participation:** The Act also says nothing about public participation as regards environmental protection.
 - There is a need to involve the citizens in environmental protection to check arbitrariness and raise awareness and empathy towards the environment.
- **Incomplete Coverage of Pollutants:** The Act does not address modern concept of pollution such as noise, overburdened transport system and radiation waves which are also an important cause for the deteriorating environment.

INDIAN FOREST ACT, 1927

This Act recognises forest dwellers' rights and makes conservation more accountable. The Act basically does two things: (i) Grants legal recognition to the rights of traditional forest dwelling communities, partially correcting the injustice caused by the forest laws, and (ii) Makes a beginning towards giving communities and the public a voice in forest and wildlife conservation.

WILDLIFE PROTECTION ACT, 1972 (WITH AMENDMENT ACTS OF 2003 AND 2006)

The act provides for the protection of wild animals, birds and plants and matters connected with them, with a view to ensure the ecological and environmental security of India. The act constitutes a National Board for Wildlife that provides guidelines for framing policies and advising Central and State Government on promotion of wildlife conservation and controlling poaching and illegal trade of wildlife and its products; Making recommendations for setting up and managing national parks, sanctuaries and other protected areas; and Suggesting measures for improvement of wildlife conservation. It also sets up National Tiger Conservation Authority. The acts sets up various provisions related to trade and penalties for hunting the animals in wild.

BIOLOGICAL DIVERSITY ACT, 2002

The government passed the biodiversity act to conserve and promote sustainable use of biological diversity and to regulate the access to biological resources of the country with equitable share in benefits. It sets up National Biodiversity Authority (NBA), State Biodiversity Board (SBB) and Biodiversity Management Committees. Besides, it aims to respect and protect knowledge of local communities

traditional knowledge related to biodiversity and secure sharing of benefits with local people as conservers of biological resources and holders of knowledge and information relating to the use of biological resources. Besides, it also has provisions for notifying heritage sites by State Government in consultation with local body.

NATIONAL GREEN TRIBUNAL ACT, 2010

The Act enables creation of a special tribunal to handle the expeditious disposal of the cases pertaining to environmental issues. The Tribunal has Original Jurisdiction on matters of “substantial question relating to environment” & “damage to environment due to specific activity.” Tribunal is competent to hear cases for several acts such as Forest (Conservation) Act, Biological Diversity Act, Environment (Protection) Act, Water & Air (Prevention & control of Pollution) Acts etc. and also have appellate jurisdiction related to above acts after establishment of Tribunal within a period of 30 days of award or order received by aggrieved party.

ROLE OF NGOs IN THE PROTECTION OF ENVIRONMENT

The protection of environment is a pressing issue. Every person, organization and Institution has an obligation and duty to protect it. Environmental protection encompasses Not only pollution but also sustainable development and conservation of natural resources and the ecosystem. Today, the necessity of environmental awareness and enforcement is more demanding and urgent than ever before.

NGOs are simply agencies or groups, which are different from government bodies. However, NGOs are distinctive in containing a voluntary component and also because they do not operate for profit. Over the past quarter of a century and especially during the past few decades there has been a rapid growth in the numbers of NGOs involved in the development, in the number of people working for NGOs and in the amount of money that flows into these voluntary agencies working in the activities such as –Disaster management and relief, development, public health, rehabilitation, environment protection etc. However, this paper focuses on the role played by NGOs particularly in the protection of environment

The emergence of NGOs represents an organized response by civil society especially in those areas in which the state has either failed to reach or done so inadequately. The importance of public awareness and NGOs involvement in environmental protection is acknowledged worldwide. NGO's have been taking a number of steps to promote discussion and debate about environmental issues, outside the broad spheres of popular media and the educational system. NGOs can make the following Contributions:

- Conducting education and citizen awareness programmes in the field of environment
- Fact – finding and analysis
- Filing public interest litigations
- Innovation and experimenting in areas which are difficult for government agencies to make changes in
- Providing expertise and policy analysis
- Providing factual and reliable information with a network of professional expert staff

- Remaining independent while passing relevant information to the public and governmental bodies
- Solidarity and support to environmental defenders
- Working in collaboration with the government for capacity building and promotion of community participation in environmental awareness and protection and
- Working out at the grass root level and reaching far – flung areas with or without the government invitation.

Achievements of environmental NGOs with special reference to

WWF (India), Teri and Clean- India

1. WWF (India) is engaged in a multitude of activities for protection and conservation of the environment in the Indian context. Climate change and energy conservation are among the chief areas of concern for the organization. The forest and biodiversity conservation division strives to promote and enhance conservation of forest ecosystems in the country through a participatory approach involving key stakeholders. Through its environment education programme, it aims at strengthening individual and institutional capacity in nature conservation and environmental protection through widespread education and awareness.

Intimately involved in the conservation of tigers in India since the late 1960s, WWF's significant efforts culminated in the launch of Project Tiger in 1973. It seeks to conserve and protect the biodiversity of maritime life and resources by sensitizing the people at large.

The pollution of the river waters and the imminent threat to aquatic life is a cause of great concern. WWF (India) has stepped in on a number of occasions to launch campaigns for securing their habitat, like 'Save the River Dolphin' project. WWF believes that if you secure an animal's habitat, you secure its right to live

2. CLEAN-INDIA

Deeply concerned with the deteriorating environmental situation in the country, Development Alternatives initiated the CLEAN-India (Community Led Environment Action Network) programme with five schools in the national capital in 1996. Today, CLEAN Delhi has about forty schools regularly involved in monitoring water and air quality in over 150 locations spread across Delhi. Over 2000 children have been directly trained on environmental assessment and improvement activities. They keep vigil, assess environmental quality, plead, cajole and lead the community in monitoring environment. Action programmes like solid waste management, plantation drives, energy conservation, paper recycling, etc. to improve local environmental conditions have also been initiated by schools, resident welfare associations, business and industrial associations as well as individual households.

Campaigns against the use of polybags, firecrackers during Diwali and toxic (chemicalbased) colours during Holi and for saving the city's 'Green Treasure' are also carried out.

The experience in Delhi indicates that when environment assessment is community based, it mobilises the community to review the local environmental conditions and take requisite measures, without waiting for undue external support.

Encouraged by the Delhi experience, NGOs from different towns have operationalised the

CLEAN-India programme. The present CLEAN-India Centers are:

Delhi, Shillong, Faizabad, Ladakh, Bangalore, Berinag (Kumaon Hills), Jhansi, Sagar, Bilaspur, Lalitpur, Madurai and Thiruvananthapuram. Almost 100 NGOs from across the country have expressed their interest to initiate the programme in their respective towns.

3. TERI (INDIA)

Tata Energy Research Institute (TERI) was formally established in 1974 with the purpose of tackling and dealing with the immense and acute problems that mankind is likely to be faced with in the years ahead:

On account of the gradual depletion of the earth's finite energy resources which are largely nonrenewable

and on account of the existing methods of their use which are polluting TERI has

launched a major project, the first phase of which is completing near completion. This project called growth with resource enhancement of Environment and nature (GREEN INDIA-2047)" has vigorously estimated the reduction in India's key natural resources during the period 1947-97, and has completed economic values of consequent losses, which in some cases are alarmingly high. On the basis of past experience and a careful analysis of the cost behind the degradation that has taken place in the past strategies for the future have been developed, Where by a fresh and creative approach can be taken in the next 50 years.

The global presence and reach attained by TERI are not only substantiated by its presence in different parts of the world, but also in terms of wide geographical relevance of its activities. Symbolic of this fact is the annual Delhi Sustainable Development Summit (DSDS), a major event focusing on sustainable development.

Human Population and the Environment

Population growth is the increase in the number of humans on Earth. For most of human history our population size was relatively stable. But with innovation and industrialization, energy, food, water, and medical care became more available and reliable. Consequently, global human population rapidly increased, and continues to do so, with dramatic impacts on global climate and ecosystems. We will need technological and social innovation to help us support the world's population as we adapt to and mitigate climate and environmental changes.

Human population growth impacts the Earth system in a variety of ways, including:

- Increasing the extraction of resources from the environment. These resources include fossil fuels (oil, gas, and coal), minerals, trees, water, and wildlife, especially in the oceans. The process of removing resources, in turn, often releases pollutants and waste that reduce air and water quality, and harm the health of humans and other species.

- Increasing the **burning of fossil fuels** for energy to generate electricity, and to power transportation (for example, cars and planes) and industrial processes.
- Increase in **freshwater use** for drinking, agriculture, recreation, and industrial processes. Freshwater is **extracted** from lakes, rivers, the ground, and man-made reservoirs.
- Increasing ecological impacts on environments. **Forests** and other habitats are disturbed or destroyed to construct urban areas including the construction of homes, businesses, and roads to accommodate growing populations. Additionally, as populations increase, more land is used for **agricultural activities** to grow crops and support livestock. This, in turn, can decrease species populations, geographic ranges, biodiversity, and alter interactions among organisms.
- **Increasing fishing and hunting**, which reduces **species populations** of the exploited species. Fishing and hunting can also indirectly increase numbers of species that are not fished or hunted if more resources become available for the species that remain in the ecosystem.
- Increasing the transport of invasive species, either intentionally or by accident, as people travel and import and export supplies. Urbanization also creates disturbed environments where **invasive species** often thrive and outcompete native species. For example, many invasive plant species thrive along strips of land next to roads and highways.
- The transmission of **diseases**. Humans living in densely populated areas can rapidly spread diseases within and among populations. Additionally, because transportation has become easier and more frequent, diseases can spread quickly to new regions.

Population Growth

Two types of population growth patterns may occur depending on specific environmental conditions:

- An exponential growth pattern (J curve) occurs in an ideal, unlimited environment
- A logistic growth pattern (S curve) occurs when environmental pressures slow the rate of growth

Exponential Growth

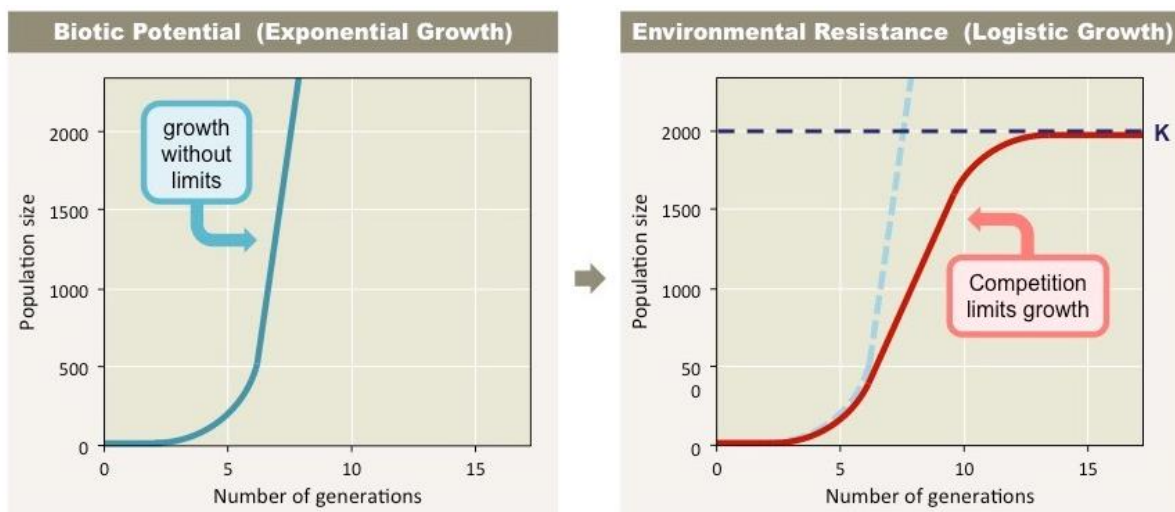
- Exponential population growth will occur in an ideal environment where resources are unlimited
- In such an environment there will be no competition to place limits on a geometric rate of growth
- Initially population growth will be slow as there is a shortage of reproducing individuals that may be widely dispersed
- As population numbers increase the rate of growth similarly increases, resulting in an exponential (J-shaped) curve
- This maximal growth rate for a given population is known as its **biotic potential**

- Exponential growth can be seen in populations that are very small or in regions that are newly colonised by a species

Logistic Growth

- Logistic population growth will occur when population numbers begin to approach a **finite carrying capacity**
- The carrying capacity is the maximum number of a species that can be sustainably supported by the environment
- As a population approaches the carrying capacity, environmental resistance occurs, slowing the rate of growth
- This results in a sigmoidal (S-shaped) growth curve that plateaus at the carrying capacity (denoted by K)
- Logistic growth will eventually be seen in any stable population occupying a fixed geographic space

Types of Population Growth



Limiting Factors

Limiting factors are environmental conditions that control the rate at which a process (e.g. population growth) can occur

- predator numbers
- availability of food and other resources
- the spread of pathogenic diseases
- weather and climate conditions,
- occurrence of natural disasters (e.g. earthquakes)

Women education and Environment

Women play a critical role in managing natural resources on family and community levels and are most affected by environmental degradation. In communities around the world, women manage water, sources for fuel, and food, as well as both forests and agricultural terrain. Women produce 60 to 80 percent of food in developing countries, while inheritance laws and local customs often prevent them from owning or leasing land and securing loans or insurance. From the high level to the grassroots, the 1992 UN Earth Summit, India's Chipko movement and Kenya's Green Belt Movement all highlighted the role of women's voices and perspectives in sustainable development.

The interdependence of women with nature can be evolved from the following observations.

1. Women are having base and skills for natural resource management.
2. Women are better managers for natural resources
3. Women have sensitiveness towards environment.
4. Women have high ecological consciousness

Women are regarded as the best caretakers of environment since ancient days.

Women play the most important role in managing environment.